INCH-POUND

MIL-PRF-27/51D
w/AMENDMENT 2
24 March 2022
SUPERSEDING
MIL-PRF-27/51D
w/AMENDMENT 1
25 March 2016

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 3 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.

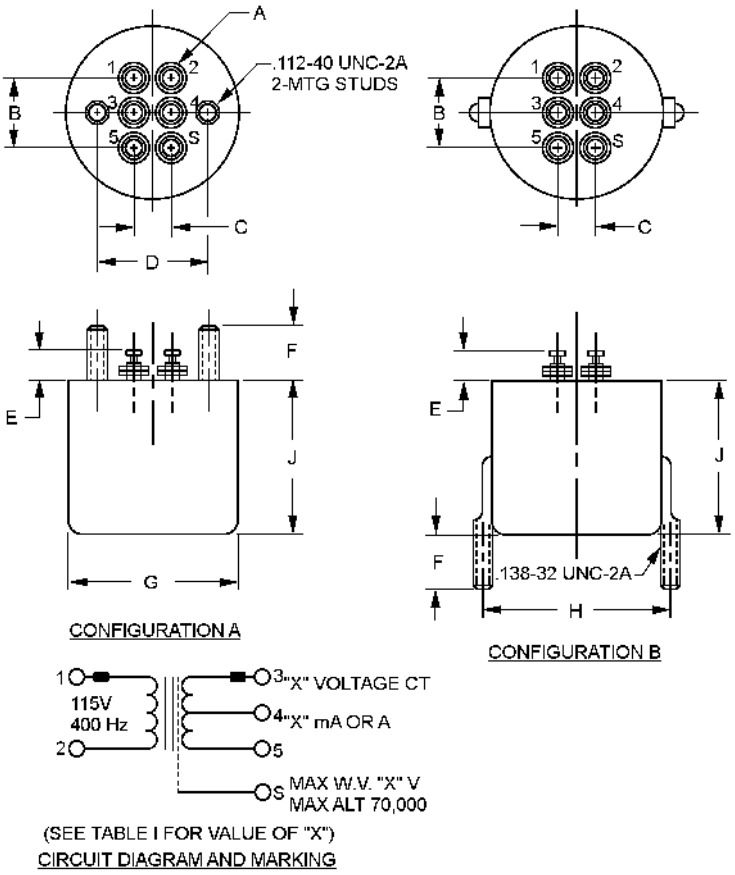


FIGURE 1. Dimensions and configuration.

AMSC N/A FSC 5950

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown. Ground terminal "S" is black.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

LTR	Inches	mm
A	.234 DIA MAX	5.94 DIA MAX
B	.500	12.70
C	.250	6.35
D	.750	19.05
E	.312 MAX	7.92 MAX
F	.344 ±.031	8.74 ±0.79
G	1.000 ±.031	25.40 ±0.79
H	1.125 ±.031	28.58 ±0.79
J	.812 ±.031	20.62 ±0.79

FIGURE 1. Dimensions and configuration – continued.

REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.7 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .312 inch (7.92 mm), maximum.

Terminal strength: MIL-STD-202-211, test condition A, 5 pounds.

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Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 6.0 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 75 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 12 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 20°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 110°C.

Vibration: MIL-STD-202-204, test condition D.

Marking location: See figure 1.

VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/51-44 shall be sufficient to grant qualification approval to M27/39-01 through M27/51-44.

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Part or Identifying Number (PIN) example:

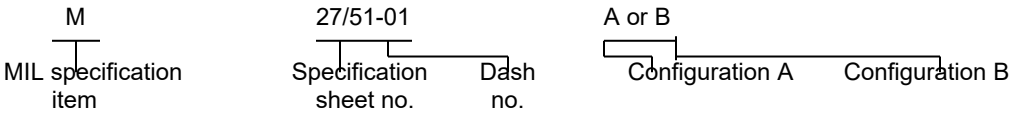


TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	6	0.500	325
02 A or B	6.3	0.476	325
03 A or B	7	0.429	325
04 A or B	7.5	0.400	325
05 A or B	8	0.375	325
06 A or B	9	0.333	325
07 A or B	10	0.300	325
08 A or B	11	0.273	325
09 A or B	12	0.250	325
10 A or B	12.6	0.238	325
11 A or B	13	0.231	325
12 A or B	15	0.200	325
13 A or B	16	0.188	325
14 A or B	18	0.167	325
15 A or B	20	0.150	325
16 A or B	22	0.136	325
17 A or B	24	0.125	325
18 A or B	26	0.115	325
19 A or B	28	0.107	325
20 A or B	30	0.100	325
21 A or B	33	0.091	325
22 A or B	36	0.083	325
23 A or B	39	0.077	325
24 A or B	43	0.070	325
25 A or B	47	0.064	325
26 A or B	51	0.059	325
27 A or B	56	0.054	325
28 A or B	62	0.048	325
29 A or B	68	0.044	325
30 A or B	75	0.040	325
31 A or B	82	0.037	325
32 A or B	91	0.033	325
33 A or B	100	0.030	325
34 A or B	110	0.027	325
35 A or B	115	0.026	325
36 A or B	120	0.025	325
37 A or B	130	0.023	325
38 A or B	150	0.020	325
39 A or B	180	0.017	325
40 A or B	200	0.015	440
41 A or B	220	0.014	440
42 A or B	240	0.013	440
43 A or B	270	0.011	580
44 A or B	300	0.010	580

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/48, MIL-PRF-27/49, MIL-PRF-27/50, MIL-PRF-27/52, MIL-PRF-27/53, MIL-PRF-27/54, MIL-PRF-27/55, MIL-PRF-27/56, MIL-PRF-27/57, MIL-PRF-27/58, MIL-PRF-27/59, MIL-PRF-27/60, MIL-PRF-27/61, MIL-PRF-27/62, MIL-PRF-27/63, MIL-PRF-27/64, MIL-PRF-27/65, MIL-PRF- 27/66, MIL-STD-202-204, MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
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Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2022-031)

Review activities:
Army – CR4, MI
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Air Force – 19

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